CSE 21 Fall 2026
Homework 3
Due date: Monday October 19, 2026 at 11:59pm

INSTRUCTIONS

This homework should be done in **groups** of one to four students, without assistance from anyone besides the instructional staff and your group members. Homework must be submitted through Gradescope by a **single representative** of your group and received by **11:59pm** on the due date. There are no exceptions to this rule.

You will be able to look at your scanned work before submitting it. You must **type** your solutions. (hand-drawn diagrams are okay.) Your group representative can resubmit your assignment as many times as you like before the deadline. Only the most recent submission will be graded.

Students should consult their textbook, class notes, lecture slides, podcasts, group members, instructors, TAs, and tutors when they need help with homework. You may ask questions about the homework in office hours, but questions on Piazza should be private, visible only to instructors.
This assignment will be graded for not only the *correctness* of your answers, but on your ability to present your ideas clearly and logically. You should explain or justify, present clearly how you arrived at your conclusions and justify the correctness of your answers with mathematically sound reasoning (unless explicitly told not to). Whether you use formal proof techniques or write a more informal argument for why something is true, your answers should always be well-supported. Your goal should be to **convince the reader** that your results and methods are sound.
KEY CONCEPTS basic probability principles, independence, conditional probability, Bayes' Theorem, Random Sampling, Random Variables, .

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1. (20 points) Consider the set H of all non-negative integer solutions to the equation:

$$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 30$$

(You can think of H as a set of 6-tuples of non-negative integers that add up to 30).

You wish to pick one of the elements of H at random and the way you decide to do it is the following:

Initialize a 6-tuple to be all zeros (0,0,0,0,0,0).

Roll a fair 6 sided die 30 times. For each roll that shows a 1, increment the 1st position, for each roll that shows a 2, increment the 2nd position and so on. (include a justification of each problem.)

   (a) What is the probability that you get the element: $(5, 5, 5, 5, 5, 5)$?

   (b) What is the probability that at least one of the entries is 0?

   (c) What is the probability that one of the entries is exactly 16?

   (d) Explain why this is not a uniform random sampling. (Show that the probability of each outcome is not equally likely.)

   (e) Which elements are the least likely to occur and why?

2. (12 points) Suppose you are at a casino and there are three slot machines. You know that two slot machines each win with probability 0.5, and the other wins with a probability of 0.9.

   You pick one of the slot machines (equally at random) and you play 8 games and win them all.

- What is the probability that you are at the slot machine that wins with probability of 0.9?
- If you decide to stay at the same slot machine (after winning 8 games), what is the probability that your next game results in a win?

3. *Partnership Dominos* is a game that involves 4 players and a *Double-6 Domino set*.

   A *Double-6 Domino set* is a set of 28 different tiles such that each tile has two values (one on each side. The values are given by a number of *pips* or dots to signify the value.) There is one domino for each combination of values from 0 through 6 (including doubles). Here is a picture of a full set of 28 Double-6 dominos:

   At the start of the round, each of the 4 players (North, South, East, West) gets 7 tiles.

   (a) What is the probability that North has the double-six tile (the tile with 6 pips on either side)?
   Solution. The probability that North has the double-six tile is:

   $$\boxed{\frac{7}{28} = \frac{1}{4}}$$

   Because there are 28 tiles total, and each player gets 7 tiles.

   (b) What is the probability that North has at least one double? (there are 7 doubles)
   Solution.
   The probability that North has at least one double is:

   $$\boxed{1 - \frac{\binom{21}{7}}{\binom{28}{7}}}$$

   Because the probability that North has at least one double is the complement of the probability that North has no doubles. 21 choose 7 is how many ways we can have doubles, and 28 choose 7 is how many combinations of tiles North can have. After dividing we get the probability of having no doubles. 1 minus that is that is the probability North has at least one double.

   (c) What is the probability that North has at least one domino with a 6 and at least one domino with a blank?
   Solution.
   Let event A be the event that North has no tiles containing a 6. Let event B be the event that North has no tiles containing a blank. Then the probability that North has at least one domino with a 6 and at least one domino with a blank is:

   $$1 - P(A) - P(B)$$

   But we are over substracting that one tile that has both a 6 and a blank. So we can add it back in by:

   $$\boxed{1 - P(A) - P(B) - P(A \cap B)}$$

   The probability of A and B is the same as part b, so out solution becomes:

   $$\boxed{1 - \frac{\binom{21}{7}}{\binom{28}{7}} - \frac{\binom{21}{7}}{\binom{28}{7}} + \frac{\binom{15}{7}}{\binom{28}{7}}} = 1 - 2 * \frac{\binom{21}{7}}{\binom{28}{7}} + \frac{\binom{15}{7}}{\binom{28}{7}}$$

   We get 15 choose 7 from the intersection of A and B = 28 - (7 + 7 - 1), because we are looking for dominos that don't have 6 or aren't blank, but we add 1 back in because of that one tile that is 6 and blank.

(d) What is the probability that North and South together hold all 7 tiles containing the value 6?
Solution.

$$\boxed{\frac{\binom{21}{7}}{\binom{28}{14}}}$$

We find the probability by finding all ways to get 7 tiles on top of the 7 fixed tiles that all contain 6. Divide that by number of ways we can pick 14 tiles and we get the probability of having all 7 tiles containing the value 6.

(e) A "Bad hand" is a set of 7 dominos with at least four doubles. What is the probability that one of the players has a bad hand?
Solution.

$$\boxed{\frac{\binom{7}{4} * \binom{21}{3} + \binom{7}{5} * \binom{21}{2} + \binom{7}{6} * \binom{21}{1} + \binom{7}{7} * \binom{21}{0}}{\binom{28}{7}}}$$

We approach every case by itself, starting with having 4 doubles (7 choose 4) and the rest is non doubles (21 choose 3). We apply this to every case of each scenario and divide by how many ways we can choose 7 tiles from 28.

4. *Fantasy Five* is a lottery game you can play for \$1. The way it works is, from the set 1,2,3,...,39, you select a subset of 5 different numbers.

The lottery commissioner also selects 5 *winning numbers* uniformly at random.

You win money based on how many of your numbers match with the *winning numbers*.

Here are the payouts:

5 out of 5 pays \$67000
4 out of 5 pays \$401
3 out of 5 pays \$15
2 out of 5 pays \$1
1 out of 5 pays \$0
0 out of 5 pays \$0

(a) (4 points) What is the expected number of matches you make with the winning numbers?
Solution.
Let X_i be the number that matches. The probability that all number match is:
$X = X_1 + X_2 + \cdots + X_5$ The expected value of that probability, because of linearity of expected values:
$E(X) = E(X_1) + E(X_2) + \cdots + E(X_5)$
The expected value of X is P(X) = $\frac{5}{39}$ So expected number of matches is:

$$\boxed{E(X) = 5 * \frac{5}{39} = \frac{25}{39}}$$

(b) (4 points) What is the expected profit you make from playing this game?
Solution.
First we need to find the probability of each payout:

$$P(0) = \frac{\binom{34}{5}}{\binom{39}{5}} = 0.483$$

$$P(1) = \frac{\binom{5}{1} * \binom{34}{4}}{\binom{39}{5}} = 0.403$$

$$P(2) = \frac{\binom{5}{2} * \binom{34}{3}}{\binom{39}{5}} = 0.104$$

$$P(3) = \frac{\binom{5}{3} * \binom{34}{2}}{\binom{39}{5}} = 0.010$$

$$P(4) = \frac{\binom{5}{4} * \binom{34}{1}}{\binom{39}{5}} = 0.0003$$

$$P(5) = \frac{1}{\binom{39}{5}} = 0.000002$$

Then expected profit is: $(0 * 0.483 + 0.403 * 0 + 0.104 * 1 + 0.010 * 15 + 0.0003 * 401 + 0.000002 * 67000) - 1 \approx \boxed{\$ - 0.50}$

(c) (4 points) What is the expected number of times you must play this game before and including your first win of at least one dollar?

Solution.

First we find the probability of getting at least one dollar = 0.11397343.
Let X be the amount of games we played to get at least 1 dollar, then:

$$E(X) = p + (1 - p) * (1 + E(X))$$

$$E(X) = 1 + (1 - p) * E(X)$$

$$E(X) = 1/p = \boxed{8.774}$$